



CHANDAN STEEL LIMITED

(GOVT. OF INDIA RECOGNISED EXPORT HOUSE)

ISO 9001 : 2015 CERTIFICATE No. 04 100 011022

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I/c. No.:

I/c. Date:

WORKS: PLOT NO. 31 TO 36, 45, 46, 48, & 49/2, 142

(EXP. AREA) GIDC, UMBERERGAON & SURVEY NO. 102/2 & 3, VILLAGE,

DEHARI, TALUKA: UMBERGAON, DIST: VALSAD, GUJARAT - 396171

GSTIN - 24AAAACC4713B1ZG



F860(824) QC 09A /00/11.01.2018

MILL TEST / INSPECTION CERTIFICATE 3.1

ACCORDING TO EN 10204 : 2005

Stappert Deutschland GmbH Willstatterstrabe 13 40549 Dusseldorf Germany	Test Certificate No.	: EXP/24-25/01039-04
	Date of Issue	: 07.10.2024
	P. O. No.	: 109904/W
	SC_No. & Date	: 24-25/0201, Dt.19.07.2024
	Invoice No. & Date	: EXP/24-25/01039, Dt.07.10.2024

PRODUCT DESCRIPTION STAINLESS STEEL HOT ROLLED ROUND BARS [HSN Code: 7222.11]	LENGTH		Net Wt.
	Ordered	Actual	2195 Kgs.
	5.0 - 6.2 Mtrs.	5.0 - 6.2 Mtrs.	

Article No.	Item No.	Heat No.	Material/Grade	Size mm	Tolerance	Process Route
-	-	CH-22200	1.4541/321 1.4878/321H UNS S32100	40.00 mm (HR)	EN 10060	Electric Melting, A.O.D. Refining, Continuous Casting, Hot Rolling

CHEMICAL COMPOSITION (Weight %)													
	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	Ti	Co	Nb	N
Min.	-	-	-	-	-	17.00	-	9.00	-	5x(C+N)	-	-	-
Max.	0.080	1.00	2.00	0.045	0.030	19.00	-	12.00	-	0.70	-	-	-
Results	0.044	0.54	1.89	0.039	0.002	17.13	0.35	9.02	0.36	0.39	0.23	0.023	0.0180

MECHANICAL PROPERTIES										
Specified Values	Test Method :- EN ISO 6892-1					Hardness EN ISO 6506-1 (BHN)	Test Method :-Charpy 'V' Notch Impact Tested (2mm) as per DIN EN ISO 148 - 1 Sample Size - 10 x 10 x 55 mm Test Temp. :- 20 °C (Joules)			
	Proof Strength N/mm² /MPa		Tensile Strength N/mm²/MPa (Rm)	Elongation after Fracture (A) % 4D / 5D	Reduction of Area (Z) %					
	0.2%(Rp0.2)	1%(Rp1.0)								
	Results	335	381	638	52 / 51	72	202-205	I	II	III

Remarks:	
1. Hot Rolled & Rough Peeled.	2. Solution annealed at 1050°C, soaked for 2-1/2 Hrs. followed by water quenching.
3. PMI test done 100% & found Satisfactory.	4. Grain size No. 5-6 as per ASTM E-112:13 Test Method.
5. Material is free from radio-active contamination.	
6. Round bar 45 mm Rolled from 200 x 200 mm Cast billets - Reduction Ratio 25.14:1.	
7. Intergranular Corrosion test carried out as per EN ISO 3651-2-1998 (Test method A) & ASTM A262 Pr. 'E' found satisfactory.	
8. The material hardness is in Accordance with NACE MR 0175/ISO 15156-1:2015 and NACE MR 0103-2015.	
9. Micro structure reveals- equiaxed grains of Austenite & free from carbide Precipitation.	
10. Material is Ultrasonically Tested as per EN 10308:2002/EN 10228-4:2016- Class-3 & found satisfactory.	
11. The material conforms to DIN EN 10088-3:2014, EN 10088-5:2009, EN 10272-2016, AD2000 W2/W10, ASTM A182/A182M-24b, A276/A276M-24a, A479/A479M-23a, A484/A484M-24, ASME SA 182/SA182 M-21, ASME SA276-21 & SA 479/SA479 M- 2021 of SECTION II PART A, & Customer Specifications TDC - TL 117 /NBR 1.4541-1.4878, Rev.1 Released in.11.2019	
12. Appoved acc. to AD 2000-Merkblatt W0 and certified acc. to PED (2014/68/EU) by Certification Body for Pressure Equipment of TUV NORD Systems (notified body, reg. no.:0045)	
13. Location of sample - Off Centre, Direction - Longitudinal, Test Temp - Room Temp., Shape of Test Piece - Round.	
14. Marking details - Company Logo, Size, Grade & Heat No.	
15. The material is manufactured /melted in Chandan Steel Ltd Mill and is of "Indian Origin".	
16. Material does not contain steel products originating in Russia mentioned in Annex XVII to the (EU) Regulation No 833/2014 and further amendment.	
17. Dimensions & Visual Inspection - Ok.	
18. No welding/weld repairs was performed on this material.	
BUNDLE NO.:	COLOUR : RED

Intended use : Building construction or civil engineering.

Regulated substance : No performance determined.

It is hereby certified that the above results are true and correct in every details and are meeting the requirements of specification & purchase order.

D.O.P. No.: 1.4541 /ROUND BAR

D.O.P. can be seen here: www.chandansteel.net

[Signature]



B. S. RAUTELA
MANAGER - Q.A./WORKS INSPECTOR